

AI Collaboration Workspace Product Requirements Document (PRD)

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1. Project Overview

AI Collaboration Workspace is an AI-powered team productivity and collaboration platform designed for developer teams, startups, and student groups. The platform combines real-time communication, task management, collaborative documentation, productivity analytics, and AI-assisted workflows into a unified workspace to reduce context switching and improve workflow continuity.

2. Problem Statement

Modern teams depend on multiple disconnected tools such as chat applications, task managers, video meeting platforms, documentation tools, and developer platforms. This creates:

- Workflow fragmentation
- Loss of productivity
- Frequent context switching
- Scattered project information
- Poor visibility into team progress

The project aims to centralize these workflows into one intelligent AI-native collaboration platform.

3. Core Objectives

- Centralize team collaboration workflows
- Provide AI-powered productivity assistance
- Enable real-time communication and collaboration
- Track productivity and contributions
- Reduce dependency on multiple external tools
- Build a scalable production-style full-stack application

4. Target Users

- Developer teams
- Student project groups
- Startups and small businesses
- Freelancers collaborating in teams
- Hackathon teams

5. Key Features

- Authentication & workspace management
- Real-time team chat and direct messaging
- Kanban-based task management
- Collaborative document editing
- AI-generated summaries and productivity insights
- Team and individual analytics dashboards
- GitHub activity and contribution tracking
- Notifications and activity feeds

6. Unique Selling Proposition (USP)

Unlike traditional collaboration platforms that simply add AI as an extra feature, this project is designed as an AI-native collaborative workspace. The AI layer understands: Team communication Task workflows Documentation Development activity Productivity patterns This enables intelligent automation, workflow optimization, and contextual productivity insights.

7. MVP Scope

Included in MVP: Authentication & authorization Workspace/team management Real-time chat system Kanban board Collaborative notes AI assistant & summaries Productivity dashboards **Excluded from MVP:** Video conferencing Mobile applications Advanced AI agents Enterprise billing systems

8. Technology Stack

Layer	Technology
Frontend	React
Styling	Tailwind CSS
Backend	Django
API Layer	Django REST Framework (DRF)
Database	PostgreSQL
Realtime Communication	Django Channels + WebSockets
Caching / Message Broker	Redis
Authentication	JWT / OAuth
AI Integration	OpenAI API
Deployment	Vercel + Render/Railway

9. System Architecture

React Frontend communicates with Django REST APIs for application logic and PostgreSQL for persistent storage. Real-time features such as messaging and notifications are powered through Django Channels, WebSockets, and Redis. AI capabilities are integrated using OpenAI APIs.

10. Functional Requirements

Users can create and join workspaces Teams can collaborate in real time Tasks can be assigned and tracked Documents support collaborative editing AI can summarize discussions and generate reports Dashboards provide productivity analytics GitHub activity can be integrated and tracked

11. Non-Functional Requirements

Responsive and modern UI Secure authentication and authorization Scalable backend architecture Efficient API performance Reliable real-time communication Clean and maintainable codebase

12. Development Roadmap

Phase 1: Project setup, authentication, database schema

Phase 2: Workspace and team system

Phase 3: Real-time communication module

Phase 4: Task management system

Phase 5: AI integration and summaries

Phase 6: Collaborative documents

Phase 7: Dashboards and analytics

Phase 8: Deployment and optimization

13. Project Vision

To build an intelligent AI-native collaboration platform that unifies communication, task management, productivity tracking, and workflow automation into one seamless system for modern teams.